

- 4 (a) (i) $\frac{1}{2}mv^2 = GMm/R$ B1
 $v^2 = 2GM/R$ A0
(ii) $g = GM/R^2$ M1
clear algebra giving $v^2 = 2gR$ A1 [3]

- (b) $\frac{1}{2}mv^2 = 3/2kT$
 $v^2 = 3kT/m$ C1
 $3kT/m = 2gR$ C1
 $T = (2 \times 6.6 \times 10^{-27} \times 9.81 \times 6.4 \times 10^6) / (1.38 \times 10^{-23} \times 3)$ C1
 $T = 2.0 \times 10^4 \text{ K}$ A1 [4]